

TOP 50

Azure Data Factory Terms for Data Engineers

*Master these key concepts to build and
manage efficient data pipelines.*

Azure Data Factory (ADF) is a powerful cloud-based ETL (Extract, Transform, Load) service. Understanding its key terms is essential for building scalable, efficient data solutions.

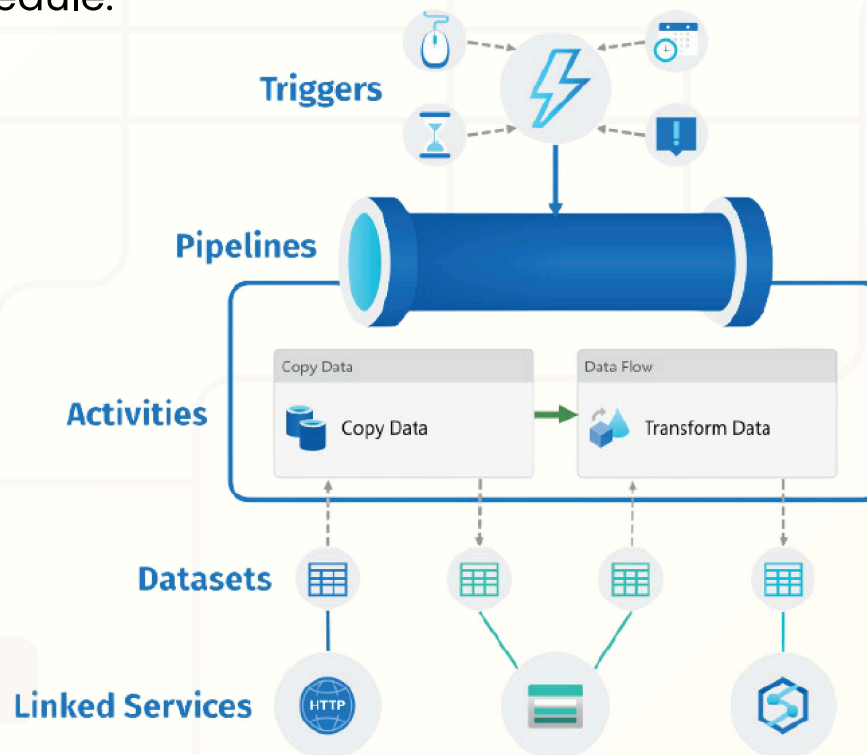


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Core Concepts

- **Pipeline:** A group of activities that perform a specific data workflow.
- **Activity:** A single step within a pipeline (e.g., data movement, transformation).
- **Dataset:** Represents data structures within data stores (e.g., files, tables).
- **Linked Service:** Defines connections to external data sources.
- **Trigger:** Initiates a pipeline run, either on-demand or on a schedule.



Data Movement

- **Copy Activity:** Transfers data from a source to a destination.
- **Sink:** The destination where processed data is written.
- **Source:** The starting point for data ingestion or processing.



- **Data Flow:** Performs complex transformations on data within ADF.
- **Integration Runtime (IR):** Provides the compute environment for activities.

Pipeline Control

- **ForEach Activity:** Executes a set of activities for each item in a collection.
- **If Condition Activity:** Executes activities based on a condition.
- **Until Activity:** Repeats a set of activities until a condition is met.

- **Wait Activity:** Pauses pipeline execution for a specified time.
- **Execute Pipeline Activity:** Calls another pipeline from the current pipeline.

Data Transformation

- **Mapping Data Flow:** Visually design data transformations at scale.
- **Wrangling Data Flow:** Perform data preparation using Power Query.
- **Derived Column:** Add new columns or update existing ones in data flow.
- **Filter Activity:** Filter data based on a condition.
- **Aggregate:** Perform aggregation operations like sum, count, or average.

Integration with Other Services

- **Azure Blob Storage:** Common data store for raw and processed data.
- **Azure SQL Database:** Cloud-based relational database used as a sink or source.



- **Azure Synapse Analytics:** Integration for large-scale analytics and data warehousing.
- **Databricks Notebook Activity:** Runs a Databricks notebook within a pipeline.
- **Power BI Activity:** Refreshes a Power BI dataset.

Data Integration

- **Hybrid Data Integration:** Connect on-premises and cloud data sources.
- **Delimited Text Dataset:** Represents  or other delimited files.
- **Parquet Dataset:** Represents  files in data pipelines.
- **JSON Dataset:** Represents  data files.
- **Query Dataset:** Use  queries to define data extraction.

Deployment and Configuration

- **ARM Templates:** Automate ADF deployments via JSON-based templates.
- **Parameter:** Dynamically pass values to pipelines or datasets.
- **Variable:** Store values within pipeline runs for temporary use.



- **Debug Parameter:** Test pipelines with different input configurations.
- **Publish:** Save pipeline changes and deploy to the live environment.

Performance and Optimization

- **Data Partitioning:** Improves performance by splitting data into partitions.
- **Parallelism:** Run multiple activities or data flows concurrently.
- **Cache Sink:** Temporarily store transformed data for reuse.
- **Throughput:** Measure of data processing speed (e.g., MB/sec).
- **Pipeline Concurrency:** The number of pipelines that can run simultaneously.

Conclusion

*Mastering these **50 terms** will help you design, build, and manage efficient data pipelines in Azure Data Factory. Start exploring and applying these concepts to optimize your data engineering workflows.*



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